

TOPIC 4.A

Structure and protection of the human integumentary system

1. What is the role of the skin in the body, and what are its main layers?
2. Name three common skin problems, and formulate the cause of their development!
3. What is the role of pigments, and where are they found?
4. What are the advantages and disadvantages of sun exposure?
5. What should be done in the event of a burn injury?

Expected knowledge	Attainable score
1. • our body's first line of defence, • epidermis, dermis, hypodermis (subcutaneous layer), • epidermis: protects against pathogens, desiccation, chemical substances, and harmful radiation; contains no blood vessels. • dermis: thermoregulation, sensation, hair follicles, sebaceous glands, sweat glands • hypodermis: nutrient storage, thermal insulation, mechanical protection	1 point 1 point 2 points 3 points 2 points
2. • Wart: a virus-induced lesion which can be protruding or flat against the skin surface. Contagious. • Blackhead, pimple (acne): caused by increased sebum production. • Fungal skin infection (dermatomycosis): fungi utilise the keratin of the epidermis, most commonly between the toes and on the nails. Contagious disease. The nail discolours, the skin itches and cracks. Requires careful treatment.	2 points 1 point 2 points
3. • they provide the colour of the skin, • they are found in the lower part of the epidermis, • their dark pigment is melanin, which absorbs the ultraviolet radiation of the Sun, thus protecting our body • upon sun exposure, pigment production increases	1 point 1 point 1 point 1 point

TOPIC 4.B

Examination of the blind spot

Investigation: Cover your right eye, then focus on the cross with your left eye and move the figure closer and further until the round spot disappears.

1. Where does the image of the cross fall on your retina when you focus on it?
2. Where does the image of the round spot fall when it disappears?
3. How do you explain what is seen?
4. Why does the blind spot not interfere with everyday vision?
5. Why is there a blind spot on the retina?

Materials, tools: figure for the examination of the blind spot.



Expected knowledge	Attainable score	Achieved score
1. On the yellow spot (macula lutea). The yellow spot is the area of sharpest vision on the retina.	2 points	
2. On the blind spot (optic disc).	2 points	
3. If we look at something with one eye, the image of the object falling on the blind spot disappears.	3 points	
4. In everyday vision, we look with two eyes, and the functioning of the cerebral cortex also helps (it visually completes the viewed image).	4 points	
5. There are no receptors located in the blind spot; the axons of the ganglion cells leave the eye here, this is the exit point of the optic nerve on the retina.	4 points	
6. Correct execution of the experiment.	5 points	
Total:	20 points	